

Integrated device for polarization-independent electro-optic modulation using a splitter-rotator structure

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An integrated device for polarization-independent electro-optic (EO) modulation based on photonic crystal is proposed. It primarily consists of three components: a polarization splitter-rotator (PSR), a combiner, and a graphene EO modulator. In the PSR, the TE₀ modes and TM₀ modes achieve polarization splitting and mode conversion. Subsequently, the two TE₀ modes pass through the combiner to merge into a single beam. Finally, the TE₀ modes enter the modulator for EO modulation. The performance parameters of the integrated device are computed and analyzed using the three-dimensional finite-difference time-domain method. The results reveal that the integrated device achieves a polarization extinction ratio of 23 dB, an insertion loss of 0.3 dB, and an extinction ratio of 21.1 dB. The modulator has a driving voltage of 3.56 V and a modulation speed of 33 GHz. The integrated device is well-suited for application in large-capacity optical communication systems. © 2024 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

Electro-optic (EO) modulators play a critical role in optical communication networks, with their performance exerting a substantial impact on the transmission capabilities of the system [1,2]. To facilitate the broader application of EO modulators, addressing polarization-dependent challenges is imperative. Due to the significant heat generated by various photon devices such as transceivers, lasers are often exposed to high-temperature environments. However, lasers cannot operate at very high temperatures (>100°C) [3,4]. One solution is to place the laser in an environment outside the transceiver that is not too harsh. This requires managing the polarization of light from the remote laser source to the input of the transceiver. The use of polarization-independent EO modulators can solve the polarization management issue. In recent years, polarization-independent EO modulators have garnered considerable attention from researchers. Materials utilized for fabricating these modulators include graphene [5–7], indium tin oxide (ITO) [8,9], and lithium niobate (LN). Bhowmik and Sikdar [10] introduced an ITO-silicon-based hybrid plasmonic waveguide modulator. The device's two electrodes operate independently, enabling the independent or synchronous modulation of TE and TM modes. The significant enhancement of the electric field within the absorption carrier accumulation layer results in a large extinction ratio (ER) (~10 dB@1550 nm). Hsu *et al.* [11] proposed a LN

polarization-independent EO modulator, employing two independent electrodes for the modulation of separate TE and TM modes. The experimental results demonstrate that the polarization-dependent loss is as low as 0.04 dB with an applied voltage of 26 V. The lengths of the two electrodes are 10 and 6 mm, leading to an overall device length of 1.6 cm. Similarly, Pan *et al.* [3] presented a LN polarization-independent EO modulator utilizing a polarization splitter-rotator (PSR) structure. This structure splits and rotates the two orthogonal linearly polarized lights of the incident light into TE₀ modes, subsequently individually modulated by two independent Mach-Zehnder modulators. The device exhibits a substantial ER (24 dB) and low insertion loss (IL) (2 dB). However, the total length of the device reaches the order of millimeters.

On the other hand, double-layer graphene structures are frequently utilized in the fabrication of polarization-independent EO modulators [12,13]. When graphene is positioned horizontally, it predominantly interacts with light in the TE modes; conversely, when graphene is vertically oriented, the interaction is mainly with light in the TM modes. Consequently, by covering graphene simultaneously in both horizontal and vertical directions on the modulator, polarization-independent EO modulation can be realized [14]. However, these modulators often encounter challenges such as a low ER and large device dimensions. Ma *et al.* [15] proposed a T-slot waveguide

modulator based on double-layer graphene, providing simultaneous support for TE and TM modes. The simulation results demonstrate that the device exhibits a low IL and compact dimensions. Nevertheless, the modulator requires a relatively high driving voltage of 7.26 V. In our previous work [16,17], a photonic crystal-based EO modulator was proposed. The modulator could only achieve the on-state and off-state modulation of the TE₀ modes. In the above research, there are still shortcomings such as low ER and large device size. Additionally, polarization-dependent issues need to be addressed.

In this work, we propose an integrated device for polarization-independent EO modulation that utilizes a PSR to convert the TM₀ modes to the TE₀ modes, followed by modulation using a graphene EO modulator. The performance parameters of the integrated device are calculated and analyzed using the three-dimensional finite-difference time-domain (3D-FDTD) method. The results indicate that the device exhibits a low IL (0.3 dB) and a high ER (21.1 dB). The modulator operates with a driving voltage of 3.56 V and achieves a modulation speed of 33 GHz. The length of the integrated device is approximately 74 μm.

2. THEORETICAL MODEL AND OPERATING PRINCIPLE

A. Theoretical Model

The theoretical model of the integrated device for polarization-independent EO modulation is illustrated in Fig. 1(a). Both TE₀ modes and TM₀ modes are simultaneously input from Port O₁. The TE₀ modes undergo no mode conversion and continue propagation along Waveguide 1. Conversely, the TM₀ modes undergo mode conversion and couple to Waveguide 2, transmitting in the form of the TE₀ modes. The two light beams arriving at Ports O₂ and O₃ represent TE₀ modes with a transmission power of 50%. When the phases of the two beams are synchronized, they coalesce at Port O₇ into a single beam with a transmission power of 100% in the TE₀ modes. Finally, the TE₀ modes enter the modulator for EO modulation.

The simplified model is depicted in Fig. 1(b). The single-polarized TM₀ modes are input from Port O₁, coupled to Waveguide 2, and transmitted to Port O₇ in the form of the TE₀ modes. S_{+i} ($i = 1 - 4$) represents the amplitude of the input wave, whereas S_{-i} ($i = 1 - 4$) denotes the amplitude of the output wave. γ_1 and γ_2 are attenuation coefficients. Following the transverse coupled mode theory, the complex amplitudes of light in Waveguide 1 and Waveguide 2 can be expressed as [18]

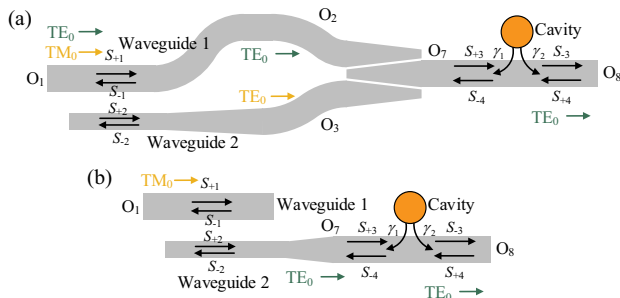


Fig. 1. (a) Theoretical model of the integrated device and (b) simplified model.

$$S_{+1} = \frac{1}{2} \left[1 - (1 + Q^2)^{-\frac{1}{2}} \right] e^{-ib_1 z} + \frac{1}{2} \left[1 + (1 + Q^2)^{-\frac{1}{2}} \right] e^{-ib_2 z}$$

$$S_{+2} = \frac{1}{2} (1 + Q^2)^{-\frac{1}{2}} (e^{-ib_1 z} - e^{-ib_2 z}), \quad (1)$$

where

$$Q = \frac{2c}{\beta_1 - \beta_2}, \quad (2)$$

$$b_{1,2} = \frac{\beta_1 + \beta_2}{2} \mp c \sqrt{1 + \left(\frac{\beta_1 - \beta_2}{2c} \right)^2} = \frac{\beta_1 + \beta_2}{2} \mp c \sqrt{1 + \frac{1}{Q^2}}, \quad (3)$$

where c is a constant. β_1 and β_2 are the propagation constants of light in Waveguide 1 and Waveguide 2, respectively. When $\beta_1 = \beta_2$, the light transmitted in Waveguide 1 can fully couple into Waveguide 2, achieving mode conversion. Therefore, to achieve optimal transverse coupling of modes, it is necessary for the propagation constants of light in both waveguides to be equal.

The modulator adopts a cavity and waveguide side coupling structure. In the off-state of the modulator, we assume that the resonant frequency of the cavity is ω_0 , and $\gamma_1 = \gamma_2 = \gamma$. The amplitude of the cavity is a . According to the temporal coupled mode theory (CMT) [16,19], the equation for the evolution of the cavity mode in time can be expressed as

$$\frac{da}{dt} = -i\omega_0 a - 2\gamma a + \sqrt{2\gamma}(S_{+3} + S_{+4}). \quad (4)$$

The relationship between the amplitudes of the input and output waves is

$$S_{-3} = S_{+3} - \sqrt{2\gamma}a, \quad (5)$$

$$S_{+3} = S_{+2}. \quad (6)$$

Therefore, the dropping efficiency $D(\omega)$ of Port O₈ is obtained as

$$D(\omega) = \left| \frac{S_{-3}}{S_{+1}} \right|^2 = \left| \frac{i(\omega_0 - \omega)}{i(\omega_0 - \omega) + 2\gamma} \cdot \frac{S_{+2}}{S_{+1}} \right|^2. \quad (7)$$

At Port O₁, $z = 0$, and $S_{+1} = 1$; that is, only incident light is input from Port O₁. Substituting Eq. (1) into Eq. (7), we can get

$$D(\omega) = \left| \frac{S_{-3}}{S_{+1}} \right|^2 = \left| \frac{i(\omega_0 - \omega) (1 + Q^2)^{-\frac{1}{2}} (e^{-ib_1 z} - e^{-ib_2 z})}{i2(\omega_0 - \omega) + 4\gamma} \right|^2. \quad (8)$$

According to the law of energy conservation, the reflection efficiency $R(\omega)$ of Port O₁ can be calculated as

$$R(\omega) = 1 - D(\omega). \quad (9)$$

Figure 2 illustrates the theoretical transmission spectrum of the integrated device for polarization-independent EO modulation when $\beta_1 = \beta_2$. At $\omega = \omega_0$, the dropping efficiency is 0, and the reflection efficiency is 100%. In other words, incident light with the resonant frequency ω is captured by the cavity and

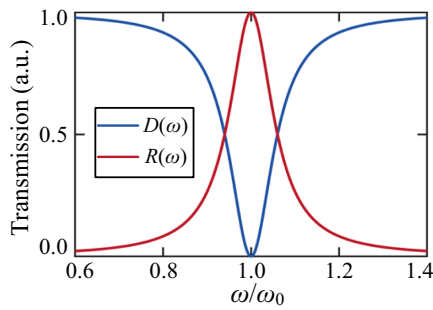


Fig. 2. Theoretical transmission spectrum of the integrated device.

entirely reflected back to the input port, placing the modulator in the off-state. The ideal operating conditions of the integrated device for polarization-independent EO modulation are obtained from the theoretical model: (1) equality of the propagation constants of light in both waveguides ($\beta_1 = \beta_2$), (2) an appropriate coupling length, (3) the modulator being in the off-state when the resonant frequency of the incident light matches the resonant frequency of the cavity ($\omega = \omega_0$), and (4) the modulator being in the on-state when the resonant frequency of the incident light is mismatched with the resonant frequency of the cavity.

B. Operating Principle

The operating principle of the integrated device for polarization-independent EO modulation is shown in Fig. 3. A, B, and C represent the PSR, combiner, and graphene EO modulator, respectively. In the PSR, both TE_0 modes and TM_0 modes are simultaneously input from Port O_1 . The TE_0 modes undergo no mode conversion and propagate along the wide waveguide to the combiner. The TM_0 modes undergo mode conversion, coupling to the etched waveguide, and then are transmitted to the combiner in the form of the TE_0 modes. In the combiner, the two TE_0 modes coalesce into the single TE_0 modes through an adiabatic taper and are transmitted to the EO modulator. In the EO modulator, when no external voltage is applied, the incident light does not couple to the cavity of the modulator and is output from Port O_8 in the form of the TE_0 modes, realizing the on-state of the modulator. When an external voltage is applied, the resonant wavelength of the modulator matches the wavelength of the incident light, causing the incident light to couple into the cavity, achieving the off-state of the modulator. Thus, the TE_0 modes and TM_0 modes input from Port O_1 are output in the form of the TE_0 modes from Port O_8 , accomplishing the functionality of polarization-independent EO modulation.

3. STRUCTURAL DESIGN AND PERFORMANCE ANALYSIS

A. Structural Design

1. Structural Design of the PSR

The three-dimensional structure of the PSR is demonstrated in Fig. 4. The device employs an asymmetric directional coupling structure. The cladding is made of SiO_2 with a refractive index of 1.44. The refractive index of the Si waveguide is 3.47. A side view along the red dashed line is shown in the inset. The width of the wide waveguide is $w_1 = 550$ nm, and the height is $h = 220$ nm. The width of the etched waveguide is $w_2 = 400$ nm. Along the y and z directions, a rectangular groove with a width of $w_3 = 230$ nm and a height of $h_1 = 120$ nm is etched. The coupling gap between the wide waveguide and the etched waveguide is $g_1 = 200$ nm, and the coupling length is $L_1 = 18$ μ m. Through the structure, the TM_0 modes in the wide waveguide can be converted into the TE_0 modes in the etched waveguide, while the TE_0 modes in the wide waveguide remain unchanged.

The steady-state field distribution of the PSR is shown in Fig. 5. In Fig. 5(a), when the light source is in the TE_0 modes, mode conversion does not occur in the coupling region due to mode mismatch. The incident light is output from Port O_2 in the form of the TE_0 modes. In Fig. 5(b), when the light source is in the TM_0 modes, the effective index of the TM_0 modes in the wide waveguide is equal to the effective index of the TE_0 modes in the etched waveguide, satisfying the mode-matching condition. The TM_0 modes are converted to the TE_0 modes and output from Port O_3 .

The effective indices of the wide waveguide and etched waveguide as a function of width are illustrated in Fig. 6. In Fig. 6(a), the effective index of the wide waveguide increases with the widening of the width w_1 . In Fig. 6(b), the effective index of the etched waveguide decreases with the increasing etch width w_3 . Due to the reduction in the cross-sectional area of the etched waveguide with the increasing w_3 , meanwhile, the effective index of the TE_0 modes in the wide waveguide is significantly larger than that of the TE_0 and TM_0 modes in the etched waveguide. Therefore, the TE_0 modes in the wide waveguide undergo no mode conversion and are output from Port O_2 in the form of the TE_0 modes. On the other hand, the effective index of the TM_0 modes in the wide waveguide is approximately equal to the effective index of the TE_0 modes in the etched waveguide, satisfying the mode-matching condition [20]. This validates the ideal operating condition (1) derived in the theoretical model. Consequently, the TM_0 modes in the wide waveguide undergo mode conversion and couple to the

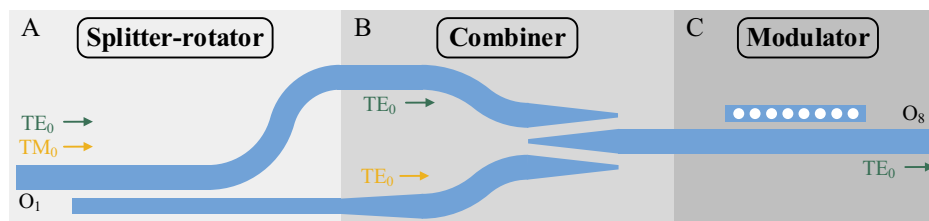


Fig. 3. Operating principle of the integrated device.

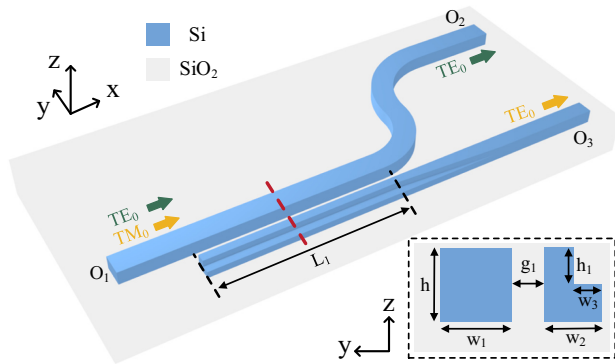


Fig. 4. Three-dimensional structural diagram of the PSR. The inset is a side view along the red dashed line.

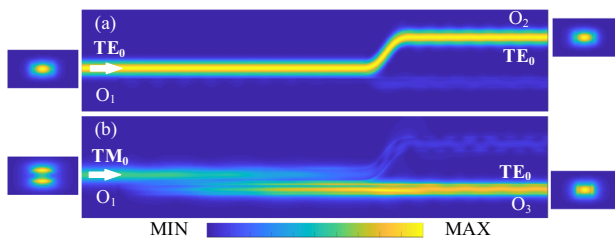


Fig. 5. Steady-state field distribution of the PSR. (a) Input in the TE_0 modes and (b) input in the TM_0 modes.

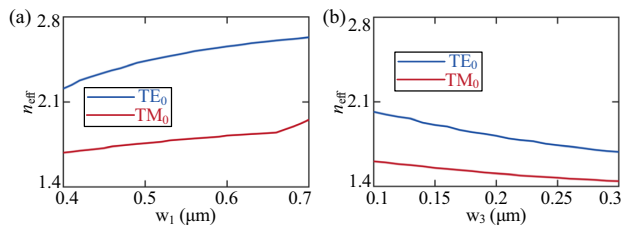


Fig. 6. (a) Effective index of the wide waveguide and (b) effective index of the etched waveguide.

etched waveguide, being output from Port O_3 in the form of the TE_0 modes.

The coupling length L_1 of the PSR affects the efficiency of mode conversion. After optimizing the coupling length L_1 , the relationship between the transmission and L_1 is visualized in Fig. 7. As L_1 increases, the transmission of the TM_0 modes first increases and then decreases because, when L_1 is either too large or too small, the TM_0 modes cannot be fully converted into the TE_0 modes. During the variation of L_1 , the transmission

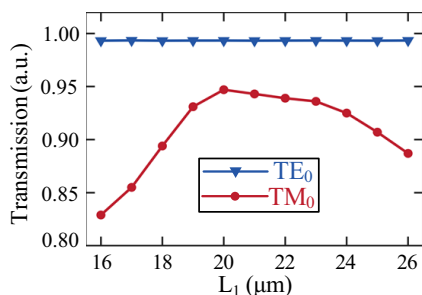


Fig. 7. Transmission versus L_1 .

of the TE_0 modes remains relatively unchanged, since the TE_0 modes do not undergo mode conversion. When L_1 is 20 μm , the transmission of the TM_0 modes is maximized. Therefore, the selected coupling length L_1 is 20 μm .

2. Structural Design of the Combiner

Researchers have proposed various types of combiners, such as Y-junctions and MMIs. Wang *et al.* [21] proposed a Y-junction combiner with an IL of less than 0.4 dB. However, the total device length reaches 95 μm , with the Y-junction itself being 30 μm long. Additionally, Yi *et al.* [22] introduced an MMI combiner with an IL of 0.21 dB at a wavelength of 1550 nm. However, the device length is 43 μm . A compact combiner is proposed, with the structural diagram shown in Fig. 8. Ports O_4 and O_5 correspond to Ports O_2 and O_3 of the PSR, respectively. The widths of the waveguides are $w_1 = 550$ nm and $w_2 = 400$ nm. The coupling region features a tapered waveguide coupling structure, comprising three identical tapered waveguides placed in reverse. The width of the waveguide varies from $w_4 = 50$ nm to $w_1 = 550$ nm. The coupling gap of the tapered waveguides is $g_2 = 50$ nm, and the coupling length is $L_2 = 5$ μm . Essentially, the principle of the combiner is also based on mode-matching, where the effective indices of the tapered waveguides are equal. The two incoming TE_0 modes from Ports O_4 and O_5 are combined into the single TE_0 modes and output from Port O_6 .

The steady-state field distribution of the combiner is illustrated in Fig. 9. Two beams of TE_0 modes combine in the coupling region to form a single beam, which is output from Port O_6 in the form of TE_0 modes.

In the combiner, the coupling gap g_2 affects the transmission when the two beams are combined into one. The variation of the transmission with g_2 is demonstrated in Fig. 10. When the gap is too large, the coupling efficiency decreases, leading to a reduction in the transmission. When g_2 is 40 nm, the transmission is maximized. Therefore, the selected coupling gap g_2 is 40 nm.

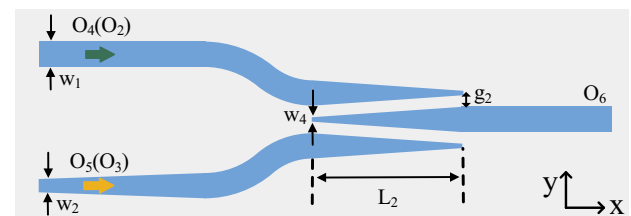


Fig. 8. Structural diagram of the combiner.

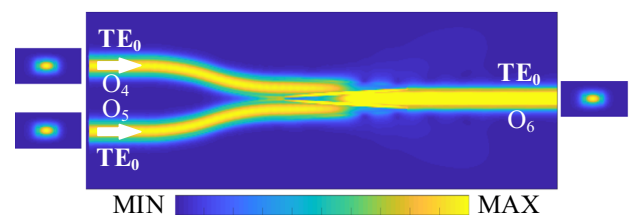


Fig. 9. Steady-state field distribution of the combiner.

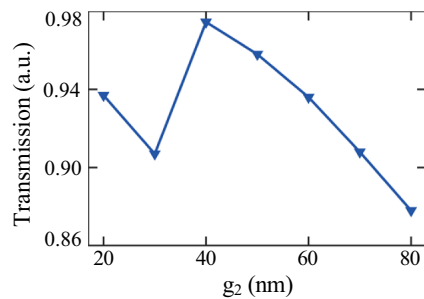


Fig. 10. Transmission versus g_2 .

3. Structural Design of the Graphene EO Modulator

Compared to traditional photonic devices, photonic crystal-based optoelectronic devices exhibit better performance and a smaller footprint [23,24]. The photonic crystal-based graphene EO modulator is visualized in Fig. 11(a). The device features a nanowire waveguide coupled with a one-dimensional photonic crystal nanobeam cavity. The side view along the red dashed line is shown in the inset. Both the waveguide and the cavity have a width of $w_1 = 550$ nm. At the center of the cavity, the distance between the waveguide and the cavity is $g_3 = 160$ nm. Meanwhile, at the position of the Bragg mirror, the distance between the waveguide and the cavity is $g_4 = 510$ nm. On the left side of the cavity, there is a 15 nm-thick silicon slab, serving as the n-doped region of the modulator with a doping concentration of $1.0 \times 10^{20} \text{ cm}^{-3}$. A layer of Al_2O_3 covers both the silicon slab and the cavity, with graphene material added on top of Al_2O_3 . The anode and cathode of the electrodes, respectively, are positioned on graphene and the silicon slab.

Figure 11(b) presents the structural diagram of the one-dimensional photonic crystal nanobeam cavity. Single-row circular holes are arranged in silicon with a lattice constant of $a = 340$ nm. In the structure, the blue region represents the Bragg mirror, with a hole radius of 108 nm. The red region signifies the photonic crystal resonant cavity area, with the hole radius gradually varying from 108 nm on both sides to 118 nm. The radii of the two black holes at the center of the cavity decrease to $r = 115.3$ nm, causing the resonant wavelength of the modulator to deviate from the target wavelength

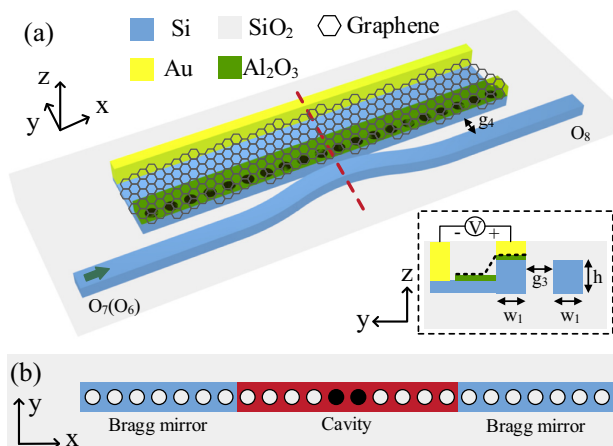


Fig. 11. (a) Three-dimensional structural diagram of the graphene EO modulator. The inset is a side view along the red dashed line. (b) Structural diagram of the nanobeam cavity.

(1550 nm). The incident light does not couple with the cavity, and the modulator is in the on-state. When a voltage is applied, the presence of graphene can significantly alter the effective index of Si [16,25]. This change in the effective index shifts the resonant wavelength of the modulator to match the target wavelength (1550 nm). In this way, all incident light is coupled into the cavity, and the modulator is in the off-state.

4. Structural Design of the Integrated Device

We integrate the PSR, combiner, and graphene EO modulator on a single chip to achieve an integrated device for polarization-independent EO modulation based on photonic crystal, as depicted in Fig. 12. The overall structure is clad with SiO_2 . The lengths of the PSR, combiner, and graphene EO modulator are 34, 16, and 24 μm , respectively. Additionally, the integration of different devices affects the resonant wavelength of the EO modulator. To achieve modulation functionality at a wavelength of 1550 nm, the hole radii of the EO modulator need slight adjustments. Specifically, the Bragg mirror area's hole radius is finely tuned to 107 nm. The photonic crystal resonant cavity area's hole radius gradually varies from 107 nm on both sides to 117 nm. The radii of the two black holes at the center of the cavity are finely tuned to 115 nm. By altering the voltage, the integrated device can achieve on-state and off-state modulation functionalities at a wavelength of 1550 nm. The structural parameters of the integrated device are summarized in Table 1.

B. Performance Analysis

The polarization extinction ratio (PER), IL, and ER are important performance parameters of the integrated device. The PER is defined as

$$\text{PER} = 10 \log(I_{\text{through}}/I_{\text{cross}}), \quad (10)$$

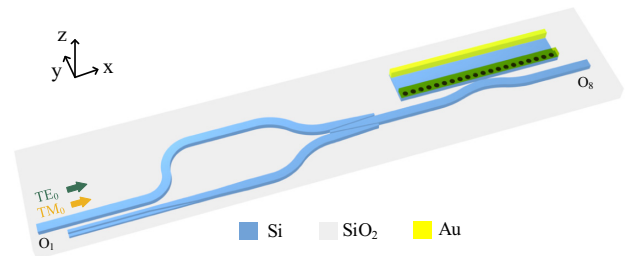


Fig. 12. Three-dimensional structural diagram of the integrated device.

Table 1. Structural Parameters of the Integrated Device

Parameter	Value	Parameter	Value
w_1	550 nm	w_2	400 nm
w_3	230 nm	w_4	50 nm
g_1	200 nm	g_2	40 nm
g_3	160 nm	g_4	510 nm
h	220 nm	h_1	120 nm
L_1	20 μm	L_2	5 μm
a	340 nm	r	115 nm

where I_{through} is the light intensity of the output port at the working wavelength, and I_{cross} is the light intensity of the non-output port at the working wavelength.

IL is defined as

$$\text{IL} = 10 \log(I_{\text{in}}/I_{\text{out}}), \quad (11)$$

where I_{in} and I_{out} are the input and output light intensity, respectively.

ER is defined as

$$\text{ER} = 10 \log(I_{\text{max}}/I_{\text{min}}), \quad (12)$$

where I_{max} and I_{min} are the output light intensity in the on-state and off-state, respectively.

1. Performance Analysis of the Integrated Device

In the graphene EO modulator, by changing the effective index of the material, the variations of the transmission and resonance wavelength are obtained, as shown in Fig. 13(a). As the effective index continuously decreases, the resonant wavelength of the modulator also decreases. However, the transmission first decreases and then increases. When $\Delta n_{\text{eff}} = -2.1 \times 10^{-3}$, the resonant wavelength of the modulator shifts to 1550 nm, the incident light couples into the cavity, and the transmission reaches a minimum (0.49%). At this point, the modulator is in the off-state. The variation of the effective index with voltage calculated using Lumerical is visualized in Fig. 13(b). When $\Delta n_{\text{eff}} = -2.1 \times 10^{-3}$, the corresponding voltage is 3.56 V. Therefore, by adjusting the voltage applied to the modulator, the on-state and off-state modulation functions can be achieved. Without applied voltage, $\Delta n_{\text{eff}} = 0$, and the modulator is in the on-state. When voltage is applied, $\Delta n_{\text{eff}} = -2.1 \times 10^{-3}$, and the modulator is in the off-state.

The performance parameters of the integrated device are analyzed using the 3D-FDTD method of commercial software Lumerical. The boundary conditions are set to PML. The mesh size is 34 nm, which is one-tenth of the lattice constant (340 nm) of the one-dimensional photonic crystal. The excitation light source is set to Mode. Both TE₀ modes and TM₀ modes are simultaneously input from Port O₁. When the modulator is in the on-state, the incident light is output from Port O₈. In the off-state, the incident light is entirely coupled into the cavity of the modulator. The transmission spectrum of the integrated device is demonstrated in Fig. 14(a). At a wavelength of 1551.4 nm, the IL of the integrated device is 0.3 dB, and the ER is 21.1 dB. The IL mainly comes from three parts, including losses in the PSR (0.13 dB), the combiner (0.11 dB), and the EO modulator (0.12 dB). Through simulation calculations, the PER of the TE₀ modes is 23 dB, and that of the TM₀ modes is

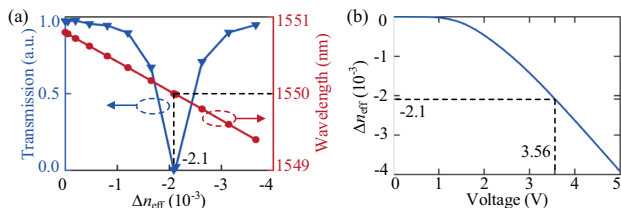


Fig. 13. (a) Transmission and wavelength versus Δn_{eff} , (b) Δn_{eff} versus voltage.

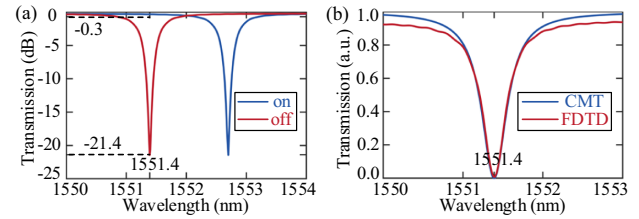


Fig. 14. (a) Transmission spectrum of the integrated device and (b) comparison of transmission spectra based on the CMT and 3D-FDTD.

18.6 dB. The comparison of transmission spectra based on the CMT and 3D-FDTD is shown in Fig. 14(b). It can be observed from the graph that the simulated transmission spectrum in the off-state is in good agreement with the theoretical transmission spectrum.

The system response time of the integrated device is visualized in Fig. 15. The system response time to reach a stable state is $T = 30$ ps, resulting in a modulation speed of $\nu = 33$ GHz for the integrated device [17,26].

Figure 16 illustrates the steady-state field distribution of the integrated device for polarization-independent EO modulation. At a wavelength of 1551.4 nm, both TE₀ modes and TM₀ modes are simultaneously input from Port O₁. The TE₀ modes undergo no mode conversion, while the TM₀ modes are converted to the TE₀ modes. The two beams of TE₀ modes are combined in the combiner and enter the graphene EO modulator. When the modulator is in the on-state, the incident light is output from Port O₈. When the modulator is in the off-state, the incident light is entirely coupled into the cavity of the modulator. By varying the voltage, the integrated device achieves polarization-independent EO modulation of both TE₀ modes and TM₀ modes.

The performance comparison between the integrated device and references is presented in Table 2. In comparison to references [9,10], the device exhibits a larger ER. Compared to

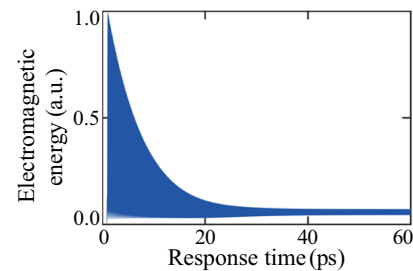


Fig. 15. System response time of the integrated device.

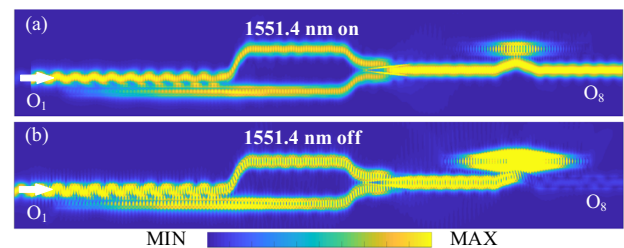


Fig. 16. Steady-state field distribution of the integrated device. (a) On-state and (b) off-state.

Table 2. Comparison of Reported Devices^a

Refs.	PER/dB	IL/dB	ER/dB	Voltage/V	Material	Method	Length
[3]	23	2	24	3.1	LN	Experiment	20 mm
[9]	—	1.4	4.5	3.5	ITO	Simulation	<24 μm
[10]	—	<1.0	~10	—5	ITO	Simulation	27 μm
[11]	>24.3	0.04	—	26	LN	Experiment	1.6 cm
[13]	—	<5.7	>17	—	Graphene	Simulation	270 μm
[15]	—	<1.0	<10	7.26	Graphene	Simulation	10 μm
This work	23	0.3	21.1	3.56	Graphene	Simulation	74 μm

^aPER, polarization extinction ratio; IL, insertion loss; ER, extinction ratio; ITO, indium tin oxide; LN, lithium niobate.

references [3,11], which also employ a PSR structure, the device achieves comparable performance with a notably shorter length of 74 μm . On the other hand, the driving voltage of the device is lower compared to that in reference [11], and the IL is lower compared to that in reference [3]. In contrast to reference [13], the device demonstrates a lower IL. Additionally, when compared to reference [15], the device operates with a lower driving voltage.

2. Tolerance Analysis of the Integrated Device

The integrated device can be fabricated on a silicon-on-insulator (SOI) wafer using deep ultraviolet (DUV) lithography processes [20,27,28]. Taking the PSR as an example, a 220 nm-thick silicon layer is deposited on a 2 μm -thick buried oxide Layer (BOX). Two masks are employed to achieve the wafer etching. The first mask is utilized for etching the 220 nm-thick silicon waveguide, while the second mask is used for etching the rectangular groove with a depth of $h_1 = 120$ nm. Finally, the unpatterned regions are fully etched down to the BOX, and an additional oxide layer is deposited as the top cladding.

In the design process of the integrated device, some structural parameters reach the nanometer scale. In the simulation environment, these structural parameters can be set to ideal values. However, during the actual manufacturing process of the device, structural parameters undergo changes due to process errors. Therefore, it is necessary to consider the impact of structural parameter errors on performance. The distance g_3 between the waveguide and the cavity and the etching width w_3 are particularly important structural parameters. The value of g_3 determines the coupling strength between the incident light and the cavity, ultimately affecting the IL and ER of the integrated device. The parameter w_3 influences the mode conversion efficiency, leading to variations in the IL and PER of the integrated device. Hence, by altering the distance g_3 between the waveguide and the cavity as well as the etching width w_3 , the variations in performance can be investigated.

The impact of the distance g_3 between the waveguide and the cavity on the IL is visualized in Fig. 17(a). As g_3 varies within the range of 70 to 250 nm, the IL remains below 0.55 dB. The variation of the ER with g_3 is shown in Fig. 17(b). With the increase of g_3 , the ER initially increases and then decreases, because distances between the waveguide and the cavity that are both too large and too small can affect the coupling efficiency between light and the cavity. To ensure an ER greater than 20 dB, g_3 can vary within the range of 100 to 220 nm. The specific values of IL and ER with varying g_3 are detailed in Table 3.

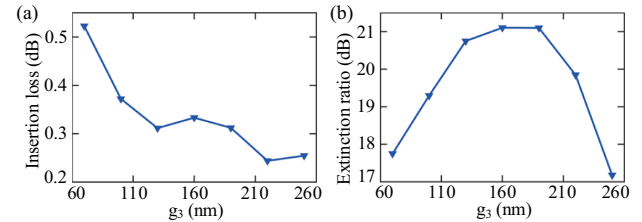


Fig. 17. (a) Insertion loss versus g_3 and (b) extinction ratio versus g_3 .

Table 3. Insertion Loss and Extinction Ratio Versus g_3

g_3/nm	Insertion Loss/dB	Extinction Ratio/dB
70	0.52	17.7
100	0.37	19.3
130	0.31	20.7
160	0.33	21.1
190	0.31	21.1
220	0.24	19.8
250	0.25	17.2

The impact of the etch width w_3 on the IL is illustrated in Fig. 18(a). It can be observed that the IL for the TE_0 modes undergoes minimal changes. For the TM_0 modes, the IL decreases initially and then increases, yet it remains below 1.0 dB throughout. Figure 18(b) depicts the influence of etch width w_3 on the PER. Similarly, the PER for the TE_0 modes exhibits minimal variation. However, the PER for the TM_0 modes is significantly affected by changes in w_3 . Due to the fact that as the etch width w_3 varies, the effective index of the TM_0 modes in the wide waveguide becomes dissimilar to that of the TE_0 modes in the etched waveguide. This disrupts the mode-matching condition, leading to a reduced efficiency in the conversion of TM_0 modes to TE_0 modes. The TE_0 modes in the wide waveguide remain unaffected by changes in w_3 , resulting in inconspicuous alterations in the IL and PER. To ensure a PER

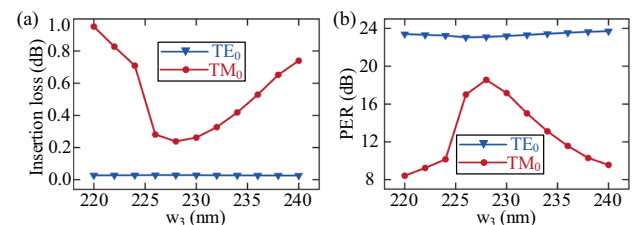


Fig. 18. (a) Insertion loss versus w_3 and (b) polarization extinction ratio versus w_3 .

Table 4. Insertion Loss and Polarization Extinction Ratio Versus w_3

w_3/nm	Insertion Loss/dB		Polarization Extinction Ratio/dB	
	TE ₀	TM ₀	TE ₀	TM ₀
220	0.027	0.95	23.4	8.4
222	0.028	0.83	23.3	9.2
224	0.028	0.71	23.2	10.2
226	0.029	0.28	23.1	17.0
228	0.029	0.24	23.1	18.6
230	0.029	0.26	23.2	17.2
232	0.028	0.33	23.3	15.0
234	0.027	0.42	23.4	13.1
236	0.027	0.53	23.5	11.6
238	0.027	0.65	23.6	10.3
240	0.026	0.74	23.7	9.6

exceeding 15 dB, the etch width w_3 can be varied within the range of 226 to 232 nm. Detailed numerical values for the IL and PER corresponding to varying w_3 are provided in Table 4.

4. CONCLUSION

An integrated device for polarization-independent EO modulation based on photonic crystal is proposed. The PSR is employed to achieve the separation and mode conversion of TE₀ modes and TM₀ modes. Subsequently, the two separated TE₀ modes are combined through a combiner and then enter the graphene EO modulator to achieve on-state and off-state modulation. The 3D-FDTD simulation results demonstrate that the device exhibits a PER of 23 dB, IL of 0.3 dB, ER of 21.1 dB, and modulation speed of 33 GHz. The device utilizes photonic crystal to design and manufacture the EO modulator, significantly reducing the size of the device. Furthermore, the integrated device employs a PSR, resulting in output light in the single-polarization TE₀ modes, which can be cascaded with wavelength division multiplexers and mode division multiplexers to meet the diverse requirements of optical communication systems. Based on the transverse coupled mode theory and the temporal coupled mode theory, the theoretical model of the integrated device is proposed. The theoretical model can guide the structural design of the integrated device. The compact structure of the integrated device facilitates easy cascading with other optical modules, making it suitable for applications in optical communication and wavelength division multiplexing systems.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

- S. Mookherjee, V. Mere, and F. Valdez, "Thin-film lithium niobate electro-optic modulators: To etch or not to etch," *Appl. Phys. Lett.* **122**, 120501 (2023).
- D. S. Zemtsov, I. A. Pshenichnyuk, S. S. Kosolobov, *et al.*, "Plasmon-assisted Si-ITO integrated electro-optical rib-shape modulator," *J. Lightwave Technol.* **41**, 6310–6314 (2023).
- Y. Pan, M. He, M. Xu, *et al.*, "Compact substrate-removed thin-film lithium niobate electro-optic modulator featuring polarization-insensitive operation," *Opt. Lett.* **47**, 1818–1821 (2022).
- A. Bigongiari, L. Giorgi, G. De Angelis, *et al.*, "Integrated remote laser source for 6G advanced antenna systems," *J. Lightwave Technol.* **40**, 519–526 (2022).
- S. W. Ye, D. Liang, R. G. Lu, *et al.*, "Polarization-independent modulator by partly tilted graphene-induced electro-absorption effect," *IEEE Photonics Technol. Lett.* **29**, 23–26 (2016).
- M. K. Shah, R. Lu, D. Peng, *et al.*, "Graphene-assisted polarization-insensitive electro-absorption optical modulator," *IEEE Trans. Nanotechnol.* **16**, 1004–1010 (2017).
- Y. Zhu, C. Deng, L. Huang, *et al.*, "Hybrid plasmonic graphene modulator with buried silicon waveguide," *Opt. Commun.* **456**, 124559 (2020).
- Y. Xu, X. Hu, Y. Dong, *et al.*, "Highly-efficient, ultra-compact and polarization-insensitive electro-absorption modulator driven by hybrid silicon-indium tin oxide-based MOS capacitors," *IEEE J. Quantum Electron.* **56**, 1–9 (2019).
- N. Dhirga, H. Mehrvar, and P. Berini, "High-speed polarization-independent plasmonic modulator on a silicon waveguide," *Opt. Express* **31**, 22481–22496 (2023).
- T. Bhowmik and D. Sikdar, "Parallel directional coupler based dual-polarization electro-absorption modulator using epsilon near-zero material," *J. Phys. D* **55**, 135107 (2021).
- C. W. Hsu, C. F. Huang, W. S. Tsai, *et al.*, "Lithium niobate polarization-independent modulator using integrated polarization splitters and mode converters," *J. Lightwave Technol.* **35**, 1663–1669 (2016).
- W. Chen, X. Fan, P. Li, *et al.*, "Polarization-insensitive electro-absorption modulator based on graphene-silicon nitride hybrid waveguide," *IEEE Photonics J.* **13**, 6600213 (2021).
- Z. Yang, R. Lu, S. Cai, *et al.*, "A CMOS-compatible and polarization-insensitive graphene optical modulator," *Opt. Commun.* **450**, 130–135 (2019).
- S. Guo, X. Li, Z. Guo, *et al.*, "Polarization-independent optoelectronic modulator based on graphene ridge structure," *Nanomaterials* **11**, 2559 (2021).
- Y. Ma, J. Li, Z. Han, *et al.*, "All-dielectric graphene-induced T-slot waveguide electro-optic modulator with polarization-independent operation," *IEEE J. Sel. Top. Quantum Electron.* **27**, 1–8 (2021).
- W. Pan, H. Chen, and Y. Hu, "Integrated device for graphene electro-optic modulation and one-dimensional photonic crystal nanobeam cavity wavelength division multiplexing," *Appl. Phys. B* **128**, 95 (2022).
- W. Pan, H. Chen, and Y. Hu, "Integrated device for electro-optic modulation and wavelength-mode division hybrid multiplexing based on graphene and photonic crystal," *Opt. Eng.* **61**, 127106 (2022).
- K. Ji, H. Chen, and Y. Hu, "An eight-channel wavelength-mode-division (de) multiplexer based on photonic crystals," *Superlattice. Microst.* **135**, 106275 (2019).
- H. Ren, C. Jiang, W. Hu, *et al.*, "Photonic crystal channel drop filter with a wavelength-selective reflection micro-cavity," *Opt. Express* **14**, 2446–2458 (2006).
- H. Guan, A. Novack, M. Streshinsky, *et al.*, "CMOS-compatible highly efficient polarization splitter and rotator based on a double-etched directional coupler," *Opt. Express* **22**, 2489–2496 (2014).
- J. Wang, B. Niu, Z. Sheng, *et al.*, "Novel ultra-broadband polarization splitter-rotator based on mode-evolution tapers and a mode-sorting asymmetric Y-junction," *Opt. Express* **22**, 13565–13571 (2014).
- Q. Yi, G. Cheng, Z. Yan, *et al.*, "Silicon MMI-based power splitter for multi-band operation at the 1.55 and 2 μm wave bands," *Opt. Lett.* **48**, 1335–1338 (2023).

23. H. M. E. Hussein, T. A. Ali, and N. H. Rafat, "A review on the techniques for building all-optical photonic crystal logic gates," *Opt. Laser Technol.* **106**, 385–397 (2018).
24. M. A. Butt, S. N. Khonina, and N. L. Kazanskiy, "Recent advances in photonic crystal optical devices: a review," *Opt. Laser Technol.* **142**, 107265 (2021).
25. W. Du, E. P. Li, and R. Hao, "Tunability analysis of a graphene-embedded ring modulator," *IEEE Photonics Technol. Lett.* **26**, 2008–2011 (2014).
26. Y. Malevich, M. S. Ergoktas, G. Bakan, *et al.*, "Video-speed graphene modulator arrays for terahertz imaging applications," *ACS Photonics* **7**, 2374–2380 (2020).
27. K. Tan, Y. Huang, G. Q. Lo, *et al.*, "Experimental realization of an O-band compact polarization splitter and rotator," *Opt. Express* **25**, 3234–3241 (2017).
28. H. Guan, A. Novack, M. Streshinsky, *et al.*, "High-efficiency low-crosstalk 1310-nm polarization splitter and rotator," *IEEE Photonics Technol. Lett.* **26**, 925–928 (2014).